Guillermo
Franco Abellán

Postdoctoral Researcher
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Personal website
GitHub: GFAbellan
InspireHEP record
ORCID

Research Interests

- Simulation-based inference
- CMB & LSS data
- Cosmic tensions
- Neutrino cosmology
- Dark sector phenomenology

Impact Snapshot

24 papers & preprints	**3700+** citations
14 h-index	**30+** talks & seminars
3 students mentored	**5** journals refereed

General Skills

Languages. Speak fluently three languages: Spanish (*native*), English (C1) and French (C1).

Programming. Expert in `Python`, `PyTorch`, `C` and `C++`, proficient in `Julia`, `JAX` and `Mathematica`.

Core tools. Contributor of public codes CLASS, CLOELIB and SWYFT; advanced user of SBI, MONTEPYTHON, COBAYA, GETDIST, SASHIMI, PYBIRD and PYLIANS.

Misc. Proficient with LaTeX, git, Slurm, Bash and HTML.

References

Vivian Poulin
LUPM, University of Montpellier
vivian.poulin@umontpellier.fr

Christoph Weniger
GRAPPA, University of Amsterdam
c.weniger@uva.nl

Bryan Zaldívar
IFIC, CSIC-University of Valencia
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Date of CV: October 2026

Education

Oct. 2019 – Oct. 2022 — **Ph.D. in Physics, specialising in Cosmology,** Université de Montpellier, France
Supervisors: Vivian Poulin and Julien Lavalle
Thesis title: *Searching for new physics with CMB precision cosmology*, available online. Date of defense: July 18, 2022

Sep. 2018 – June 2019 — **M.Sc. in Astroparticle Physics and Cosmology,** Université de Montpellier, France
Top grade in my graduating class (17.7/20)
Master internship: *Signatures of dark matter in the CMB*

Sep. 2013 – June 2018 — **B.Sc. in Physics,** Universidad de Murcia, Spain and Universität Ulm, Germany (Exchange).
Top grade in my graduating class (8.1/10); Best Graduate Award
Collaboration Scholarship (€2k competitive research grant)
Bachelor's thesis: *An introduction to supersymmetry*

Academic positions

Nov. 2025 – current — **Postdoctoral Researcher** at Institut de Física Corpuscular (IFIC), CSIC-Universitat de València, Spain

Nov. 2022 – Sep. 2025 — **Postdoctoral Researcher** at GRAPPA Institute, University of Amsterdam, The Netherlands

Research summary: Theoretical cosmologist connecting particle physics models (dark sector, neutrinos) with high-precision data (CMB, LSS). During my PhD, I focused on the model side, deriving novel constraints on various ΛCDM extensions motivated by cosmic tensions. As a postdoc, I focus on the methods side, building state-of-the-art statistical inference tools for complex problems in cosmology.

Research Output and Visibility

- **24 papers** (3 collaboration, 3 preprints); **lead author on 8** and second author on 9. Over 3700 citations (1500+ excluding collaboration papers) and an h-index of 14. Published in high-impact journals including JCAP, JHEP, PRD, MNRAS and Physics Reports. Full publication list can be found here.
- PRD Editor's Suggestion (single-author article), featured in APS *Physics* Magazine.
- **30+ conference talks and seminars** at international research institutions, including 4 invited conference talks. Selected talks, including slides, can be found here.
- **Public Codes**: Developer of SIREENI, SWYFT-LSS (SBI codes) and four non-trivial extensions of the Boltzmann solver CLASS, see GitHub for more details.

Leadership, Service and Recognition

- Member of the **Euclid Consortium** since 2023; **paper lead** on a Key Project for testing decaying DM with Euclid DR1 data; contributor to the CLOELIB library.
- **Referee** in leading journals (A&A, JCAP, Nature Astronomy, OJA and EPJ C).
- Organizer of bi-weekly *Amsterdark* seminar series (1 year); LOC member for EuCAIFCon 2024; co-organizer of the ISAPP PhD school *Exploring the Dark Universe* (2023).
- Handled GRAPPA's official social media accounts (2022–2025); gave a crash course on the CLASS code (2023).
- **Shortlisted** for CNRS competition 2024 (section 2).

Mentoring, Teaching and Outreach

- **Mentoring**. Supervised 1 Master student (N. Terzaghi) and co-supervised 2 PhD students (F. Zimmer, O. Savchenko), leading to three publications.
- **Teaching**. Tutored *Fundamentals of Physics for Biologists* (Undergrad course, 64h); teaching assistant for 2 Master's courses: *Student Seminar Theoretical Physics* (24h) and *Academic Skills for Research* (40h).
- **4 public talks**: *Pint of Science Festival*, Valencia (2026); Journée des chercheurs non-permanents, LUPM Montpellier (2022); invited talk for physics undergraduates, U. Murcia (2021); *PhD Pub*, Montpellier (2019).